

Day 1: Environment Setup

Learning Objectives

By the end of today, you will:

- Install Python and all required computer vision packages
- Verify each installation works correctly
- Set up your development environment for the entire 30-day course

Part 1: Theory - Understanding the Tools

What Are We Installing?

Before we start installing, let's understand what each package does:

Package	Purpose
Python	The programming language we'll use throughout this course
OpenCV (cv2)	The main computer vision library - handles image loading, processing, and display
NumPy	Provides multi-dimensional array support - images are represented as NumPy arrays
Matplotlib	Plotting library - we'll use it to display histograms and images
scikit-image	Additional image processing algorithms
Mahotas	Complementary image processing functions
scikit-learn	Machine learning library - used for classification tasks

Important Note on Python & OpenCV Versions

OpenCV 3.0+ supports Python 3! All code in this course works with both Python 2.7 and Python 3+. We recommend using Python 3.6 or higher.

How Images Become Numbers

In computer vision, images are represented as grids of numbers. Each number represents the **intensity** (brightness) of a pixel:

- **Grayscale image:** Single value per pixel (0 = black, 255 = white)
- **Color image (RGB):** Three values per pixel (Red, Green, Blue)
- **OpenCV quirk:** Stores colors in **BGR** order, not RGB!

Example: A pure red pixel in OpenCV = (0, 0, 255) not (255, 0, 0)

Part 2: Installation Instructions

Step 1: Install Python

For Windows:

1. Go to <https://www.python.org/downloads/>
2. Download Python 3.8 or higher
3. **IMPORTANT:** Check "Add Python to PATH" during installation
4. Verify installation: Open Command Prompt and type:

```
python --version
```

For macOS:

```
brew install python3
```

Or download from [python.org](https://www.python.org)

For Linux (Ubuntu/Debian):

```
sudo apt-get update  
sudo apt-get install python3 python3-pip
```

Step 2: Install pip (Package Manager)

pip usually comes with Python. Verify it works:

```
pip --version
```

or

Step 3: Install NumPy and SciPy

```
pip install numpy scipy
```

Why these first? OpenCV and other libraries depend on NumPy.

Step 4: Install Matplotlib

```
pip install matplotlib
```

Step 5: Install OpenCV

```
pip install opencv-python
```

This installs the main OpenCV package. For additional contrib modules:

```
pip install opencv-contrib-python
```

Step 6: Install scikit-image

```
pip install scikit-image
```

Step 7: Install Mahotas

```
pip install mahotas
```

Step 8: Install scikit-learn

```
pip install scikit-learn
```

Optional: Install Jupyter Notebook (Recommended)

```
pip install jupyter
```

This allows you to run code in interactive notebooks.

Part 3: Verification Script

Create a file called “verify_install.py”. You can use any text editor (**Notepad, VS Code, PyCharm, etc.**).

We recommend using **PyCharm** or **VS Code**

```
# verify install.py
# Run this script to confirm all packages installed correctly

from __future__ import print_function
import sys

print("=" * 50)
print("Python Version Verification")
print("=" * 50)
print("Python version:", sys.version)
print("")

# Test NumPy
try:
    import numpy as np
    print("[SUCCESS] NumPy version:", np.__version__)
except ImportError:
    print("[FAILED] NumPy not installed")

# Test OpenCV
try:
    import cv2
    print("[SUCCESS] OpenCV version:", cv2.__version__)
except ImportError:
    print("[FAILED] OpenCV not installed")

# Test Matplotlib
try:
    import matplotlib
    print("[SUCCESS] Matplotlib version:", matplotlib.__version__)
except ImportError:
    print("[FAILED] Matplotlib not installed")

# Test scikit-image
try:
    import skimage
    print("[SUCCESS] scikit-image version:", skimage.__version__)
except ImportError:
    print("[FAILED] scikit-image not installed")

# Test Mahotas
try:
    import mahotas
    print("[SUCCESS] Mahotas version:", mahotas.__version__)
except ImportError:
    print("[FAILED] Mahotas not installed")

# Test scikit-learn
try:
    import sklearn
    print("[SUCCESS] scikit-learn version:", sklearn.__version__)
except ImportError:
    print("[FAILED] scikit-learn not installed")

print("=" * 50)
print("Verification Complete!")
```

Run the verification script:

```
python verify_install.py
```

Expected output: All packages should show "[SUCCESS]" with their version numbers.

Part 4: Create Your First Image (Bonus)

If all installations worked, let's create a simple image from scratch:

```
# first_image.py
# Creates a blank image and displays it

import numpy as np
import cv2

# Create a 300x300 pixel black image (3 channels for color)
# dtype="uint8" means each value is 0-255
image = np.zeros((300, 300, 3), dtype="uint8")

# Display the image
cv2.imshow("My First Image", image)

# Wait for key press (0 means wait indefinitely)
cv2.waitKey(0)

# Close all windows
cv2.destroyAllWindows()
```

Run it:

```
python first_image.py
```

You should see a black window. Press any key to close it.

Part 5: Troubleshooting Common Issues

Problem: "pip not recognized"

Solution: Python wasn't added to PATH. Reinstall Python and check "Add to PATH"

Problem: OpenCV import error

Solution: Try

```
pip install opencv-python --upgrade
```

Problem: Permission errors on macOS/Linux

Solution: Use

```
sudo pip install package_name
```

Problem: "module not found" after installation

Solution: You might have multiple Python versions.

Try

```
python3 -m pip install package_name
```

Day 1 Checklist

- Python installed (check with `python --version`)
- pip working (check with `pip --version`)
- NumPy installed
- OpenCV installed
- Matplotlib installed
- scikit-image installed
- Mahotas installed
- scikit-learn installed
- Verification script runs without errors
- First image displays successfully

Quick Reference: Essential pip Commands (For understanding purpose only)

```
# Install a package
pip install package_name

# Uninstall a package
pip uninstall package_name

# List all installed packages
pip list

# Upgrade a package
pip install --upgrade package_name

# Install specific version
pip install package_name==1.2.3
```

Self-Assessment Questions

1. What does OpenCV stand for?

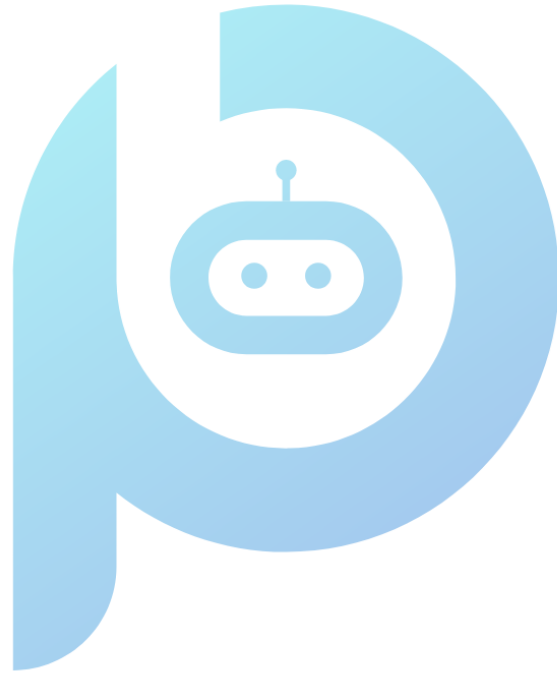
2. Why do we use NumPy in computer vision?
3. What is the range of values for a pixel in an 8-bit image?
4. What color order does OpenCV use (RGB or BGR)?
5. What does `cv2.waitKey(0)` do?

Answers:

1. Open Source Computer Vision Library
2. Images are represented as NumPy arrays for efficient processing
3. 0 to 255
4. BGR (Blue, Green, Red)
5. Waits indefinitely for a key press before continuing

Congratulations! You've completed Day 1

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